

Truly Supercritical Trade-offs for Resolution, Cutting Planes, Monotone Circuits and Weisfeiler–Leman

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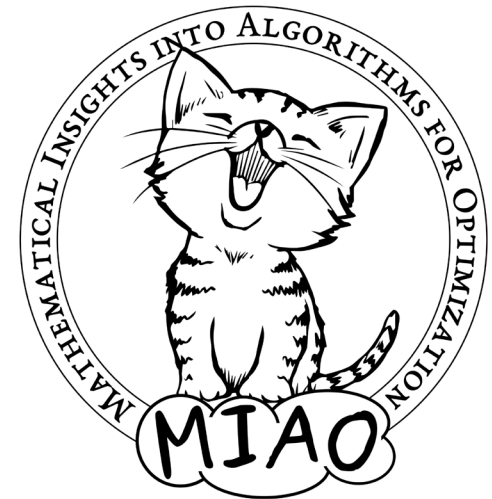


Duri Andrea
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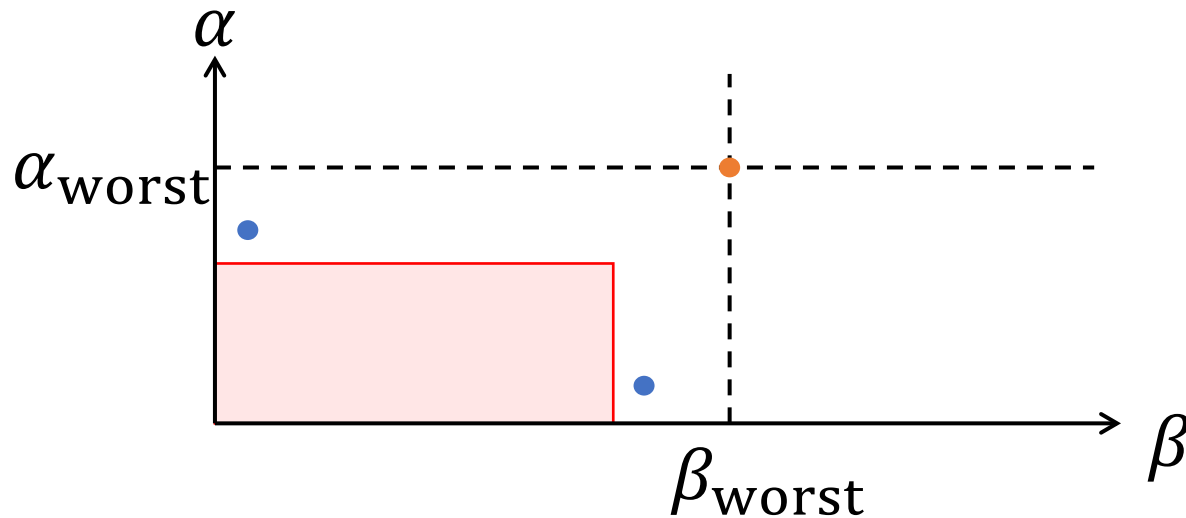


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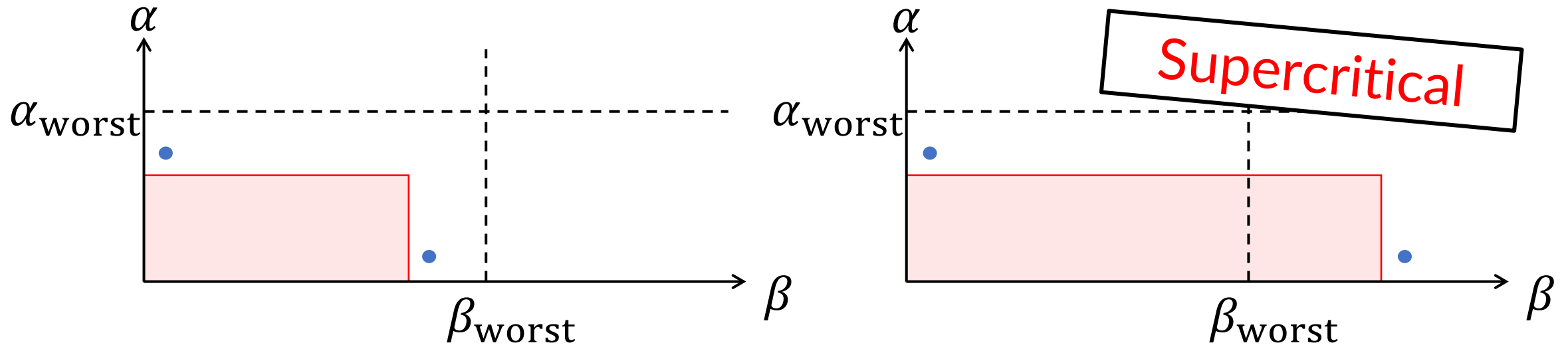
What Is a Trade-off Result?



Computational model with two complexity measures α, β (e.g. α = time and β = space)

- brute force algorithm can achieve worst case
- can optimize β , but then α bad
- can optimize α , but then β bad
- impossible to optimize both

A New Kind of Trade-off [Razborov '16]

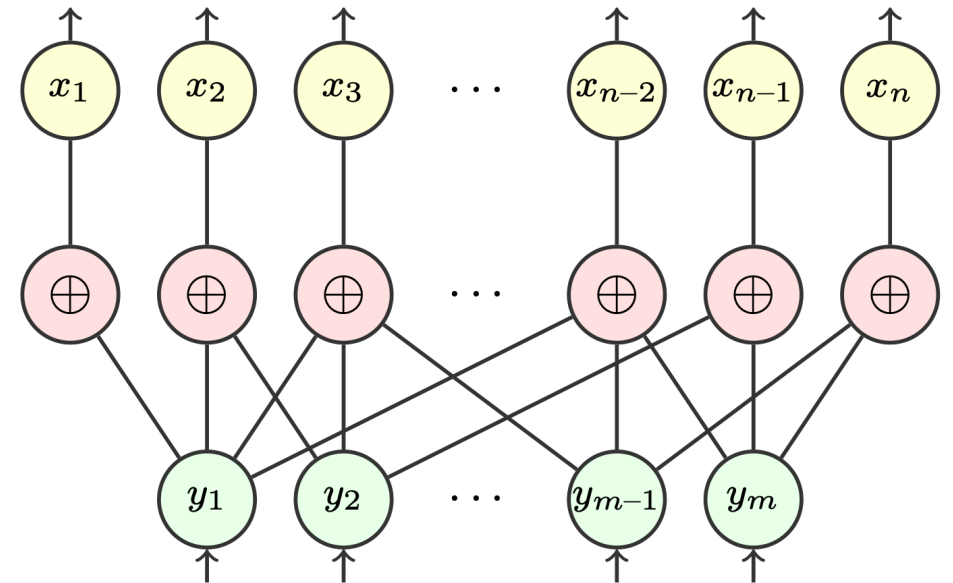


→ Optimizing α pushes β way beyond brute-force worst case

Follow-up work in [Raz'17, Raz'18, BN'20, FPR'22, BN'23, GLN'23, BT'24, CD'24]

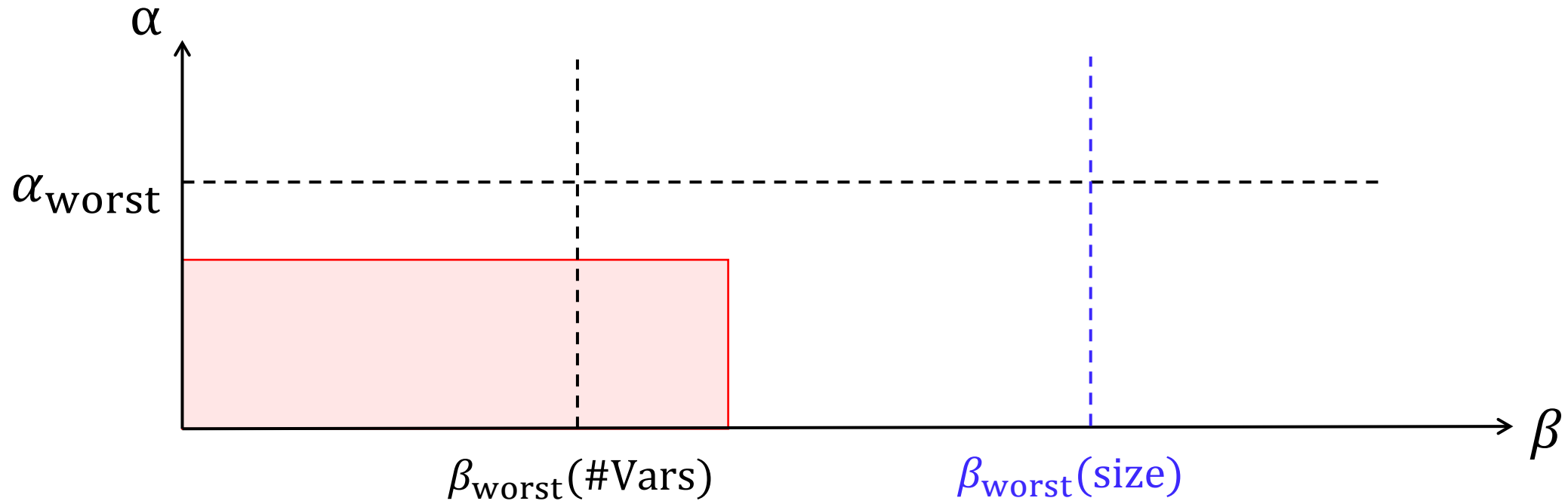
Supercritical Trade-offs Through Hardness Condensation

- Take medium-hard input in variables $x_1, \dots, x_n$
- «Condense» by substituting x_i by XORs over distinct subsets of variables $y_1, \dots, y_m$
- Show hardness is nearly preserved
- But measured in $m \ll n$: **supercritical**



Used in [Razborov '16, BN'20, FPR'22, BN'23, GLN'23, CD'24, ...]

Supercritical in What?



All trade-offs supercritical in # variables only, except [Berkholz '12, BBI '12, BNT '13]

→ Can we prove similar trade-offs *truly* supercritical *in input size*?

Overview of Our Results (Informal)

Truly supercritical trade-offs for

- depth vs width in resolution
- size vs width in tree-like resolution
- size vs depth in resolution and cutting planes
- size vs depth for monotone circuits
- dimension vs iteration number for Weisfeiler–Leman

Answering open questions in [Razborov '16, GGKS '18, FGIPRTW '21, FPR '22, GLNS '23]

Related concurrent work by [Göös-Maystre-Risse-Sokolov '25]

Next up:

- More precise statement of the results
- Preliminaries to make sense of statements

Resolution Proof System

Goal: prove CNF formula unsatisfiable
Proof of unsatisfiability: Refutation

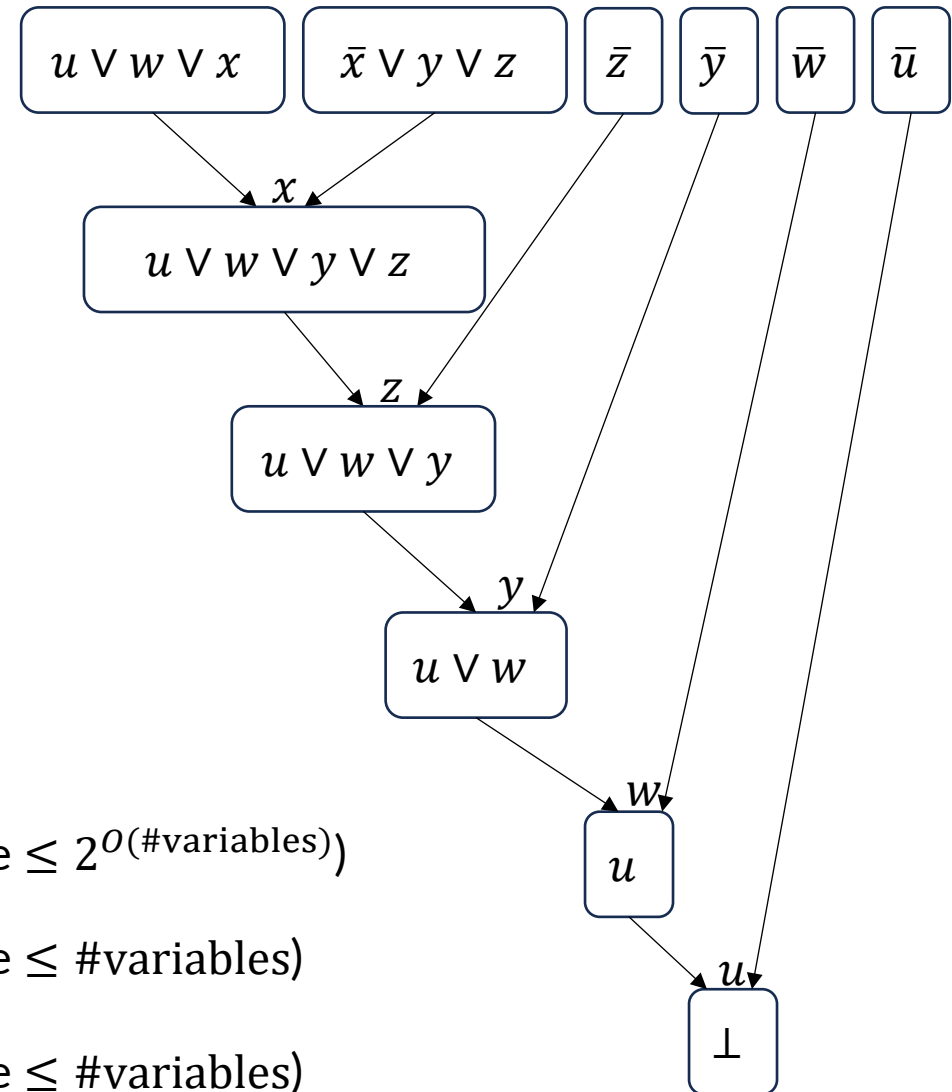
Resolution rule:

$$\frac{C \vee x \quad D \vee \bar{x}}{C \vee D}$$

size = #nodes = 11 (worst case $\leq 2^{O(\text{\#variables})}$)

width = max clause size = 4 (worst case $\leq \text{\#variables}$)

depth = max path length = 5 (worst case $\leq \text{\#variables}$)



Our Results: Resolution

Theorem (depth-width trade-off for resolution)

- $\exists$ CNF formulas F_n on n variables s.t.
- Refuted by resolution in width $w = \text{poly}(\log(n))$
 - But width $\leq 1.9w \Rightarrow$ supercritical depth superlinear($|F|$)

Theorem (size-depth trade-off for resolution)

- $\exists$ CNF formulas F_n on n variables s.t.
- Refuted by resolution in size $s = \text{quasipoly}(|F|)$
 - But size $\leq ns \Rightarrow$ supercritical depth superpoly($|F|$)

Theorem (size-width trade-off for tree-like resolution)

- $\exists$ CNF formulas F_n on n variables s.t.
- Refuted by tree-like resolution in width $w = \text{poly}(\log(n))$
 - But width $\leq w + \sqrt{w} \Rightarrow$ supercritical size $\exp(\text{superpoly}(|F|))$

All results supercritical in input size

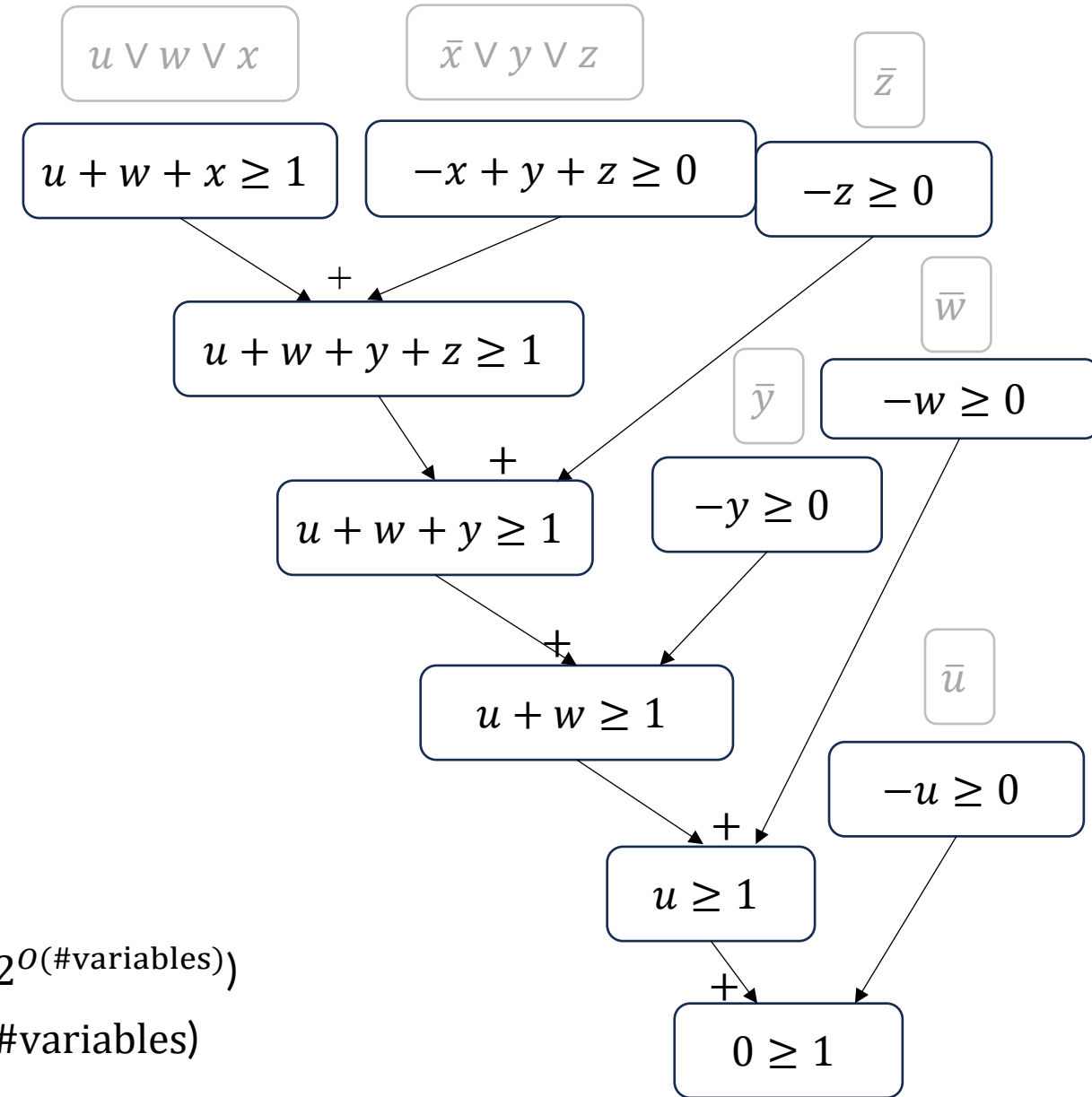
Cutting Planes Proof System

- Translate clause $\bar{x} \vee y \vee z$ to linear inequality $(1 - x) + y + z \geq 1$

Derivation rules:

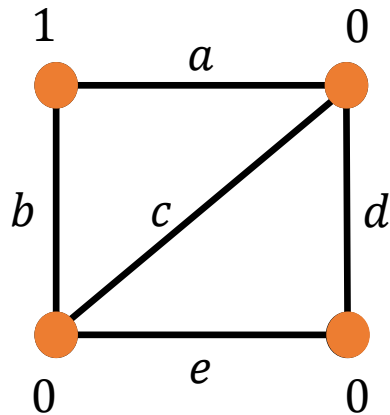
- Addition:
$$\frac{\sum a_i x_i \geq A, \quad \sum b_i x_i \geq B}{\sum a_i x_i + \sum b_i x_i \geq A + B}$$
- Multiplication:
$$\frac{\sum a_i x_i \geq A}{\sum c a_i x_i \geq cA}, \quad c \in \mathbf{N}^+$$
- Division:
$$\frac{\sum c a_i x_i \geq A}{\sum a_i x_i \geq \lceil A/c \rceil}, \quad c \in \mathbf{N}^+$$

size = #nodes (worst case $\leq 2^{O(\text{\#variables})}$)
 depth = max path length (worst case $\leq \text{\#variables}$)



Tseitin Formulas (Handshake Lemma)

- Graph $G = (V, E)$
- Labelling $\text{lbl}(v) \in \{0, 1\}$ for $v \in V$ s.t. $\sum_v \text{lbl}(v)$ odd
- Edges $e \in E \Leftrightarrow$ variables x_e
- Constraints $\sum_{e \ni v} x_e \equiv \text{lbl}(v) \pmod{2}$



$$\begin{array}{l} x_a \vee x_b \\ \overline{x_a} \vee \overline{x_b} \end{array}$$

$$\begin{array}{l} \overline{x_a} \vee \overline{x_b} \vee \overline{x_d} \\ \overline{x_a} \vee x_b \vee x_d \\ x_a \vee \overline{x_b} \vee x_d \\ x_a \vee x_b \vee \overline{x_d} \end{array}$$

$$\begin{array}{l} \overline{x_b} \vee \overline{x_c} \vee \overline{x_e} \\ \overline{x_b} \vee x_c \vee x_e \\ x_b \vee \overline{x_c} \vee x_e \\ x_b \vee x_c \vee \overline{x_e} \end{array}$$

$$\begin{array}{l} x_d \vee \overline{x_e} \\ \overline{x_d} \vee x_e \end{array}$$

- Extensive study since [Tseitin '68]
- Exponential resolution size lower bound [Urquhart '87]
- Also studied for sum-of-squares (SoS), bounded-depth Frege, stabbing planes, ...

Supercritical trade-offs for Tseitin Formulas?

- Long-standing conjecture: Tseitin formulas exponentially hard for cutting planes **FALSE!**

Theorem [Dadush, Tiwari '20]

Exist size- $n^{O(\log n)}$ cutting planes proofs for Tseitin

- Cutting planes requires depth $\geq \Omega(n)$ [FGIPRTW '21]
- But [Dadush, Tiwari '20] refutations have supercritical depth $n^{O(\log n)}$

Do Tseitin formulas yield supercritical trade-offs for size vs depth in cutting planes?

- Supercritical trade-offs in # variables (though not for Tseitin) [Fleming-Pitassi-Robere '22]

Our Result: Cutting Planes

Theorem (size-depth trade-off for cutting planes)

$\exists$ CNF formulas F_n on n variables s.t.

- Refuted by cutting planes in size $s = \text{quasipoly}(|F|)$
- But size $\leq ns \Rightarrow$ supercritical depth $\text{superpoly}(|F|)$

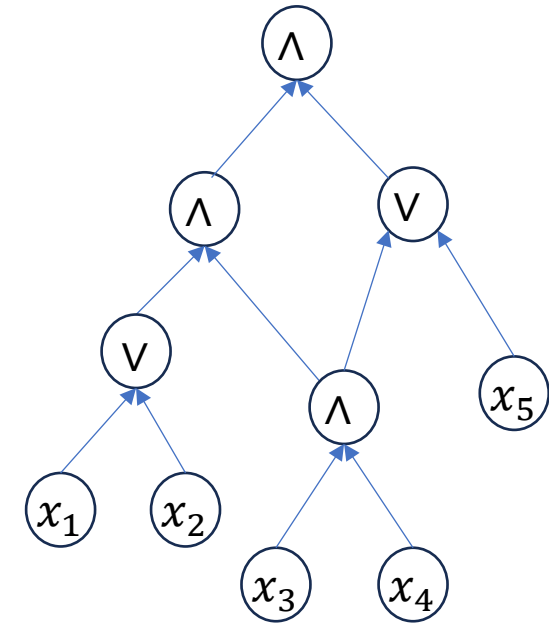
Supercritical
in input size

- Does not resolve question for Tseitin formulas
- But Tseitin formulas used to construct formulas with provable trade-offs

Monotone Circuits

- Boolean circuits with AND, OR gates
- Compute Boolean functions $f: \{0,1\}^n \rightarrow \{0,1\}$

size = #nodes (worst case $\leq 2^{O(n)}$)
depth = max path length (worst case $\leq n$)



Previous work:

Non-supercritical trade-offs: size $O(n) \Rightarrow$ depth $\Omega(n/\text{poly}(\log(n)))$

[KW '90, RM '97, GP '14, dRMNPRV '20]

Our Result: Monotone Circuits

Theorem (supercritical trade-off for monotone circuits)

$\exists$ monotone functions f_n on n variables s.t.

- Computed by monotone circuit of size $s = \text{quasipoly}(n)$
- But size $\leq ns \Rightarrow$ supercritical depth $\text{superpoly}(n)$

Proof Structure

Proof in two steps:

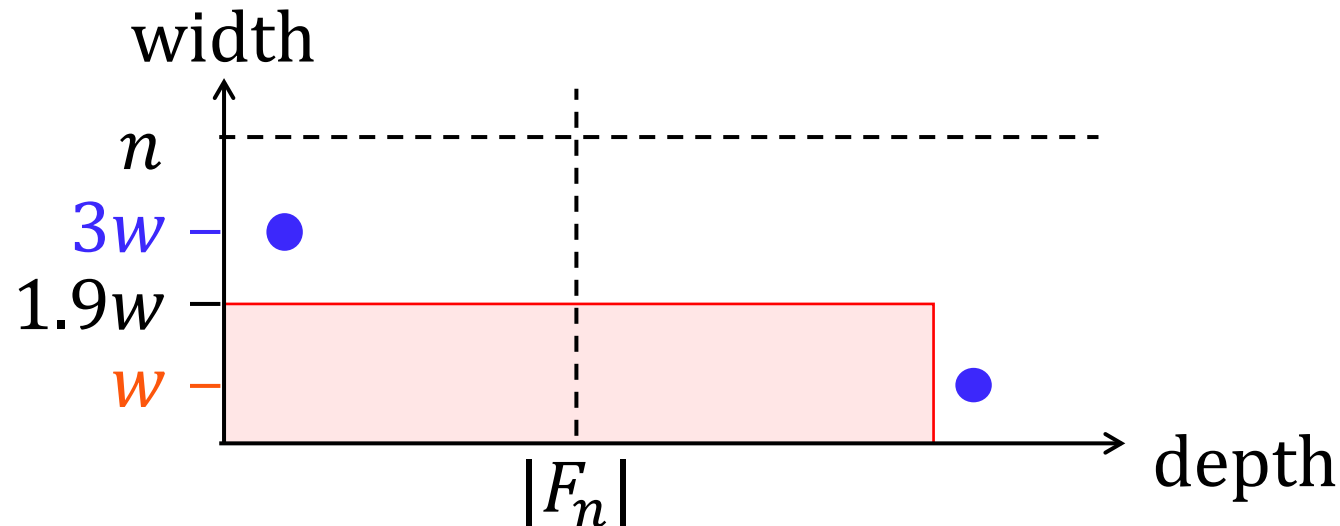
1. Establish width vs depth trade-off result for resolution
2. Derive other proof & circuit complexity trade-off results with lifting

Resolution Width-Depth Trade-off

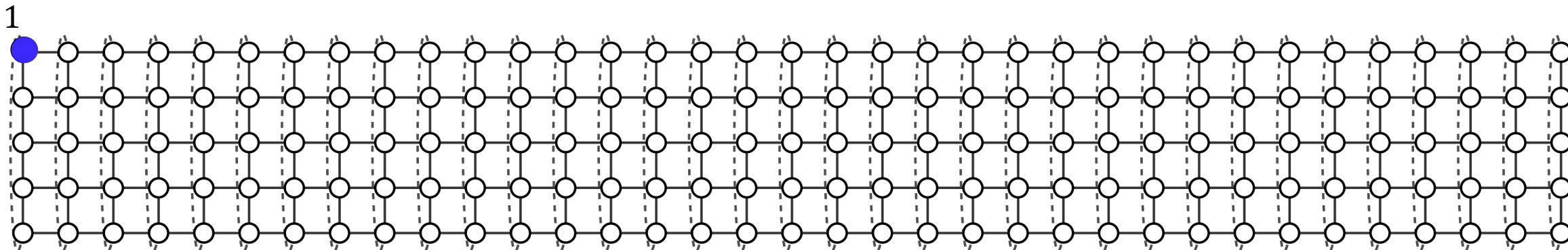
Theorem

For any $c < k < \frac{n}{2 \ln n}$ there are 4-CNF formulas s.t.

- formula size s ($\approx n^c$)
- exists proof in width $k + 3$
- but width $< k + c \Rightarrow$ depth $> s^{k/c}$



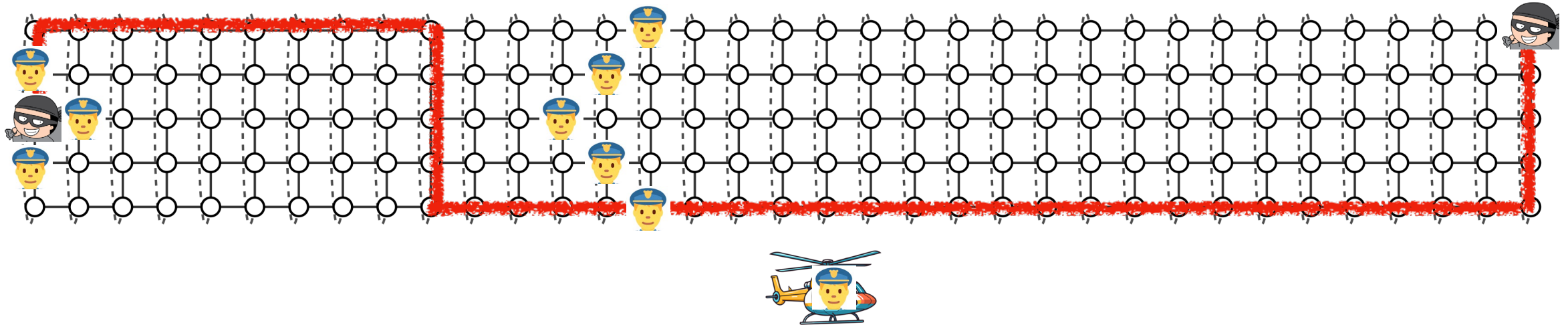
Starting Point: Tseitin Formula on Cylinder Graph



- Long, skinny cylinder $G = (V, E)$
(wrap-around vertically, but not horizontally)
- Vertex at top left labelled 1, all other vertices labelled 0
- Edges $e \in E \Leftrightarrow$ variables x_e
- Constraints: $\sum_{e \ni v} x_e \equiv \text{label}(v) \pmod{2}$
 $\Rightarrow$ contradictory due to Handshake Lemma

$$\begin{array}{l}
 x_{\text{up}} \vee x_{\text{down}} \vee x_{\text{right}} \\
 x_{\text{up}} \vee \overline{x_{\text{down}}} \vee \overline{x_{\text{right}}} \\
 \overline{x_{\text{up}}} \vee x_{\text{down}} \vee \overline{x_{\text{right}}} \\
 \overline{x_{\text{up}}} \vee \overline{x_{\text{down}}} \vee x_{\text{right}}
 \end{array}$$

Trade-offs by Analyzing the Cop-Robber Game [Seymour-Thomas '93]

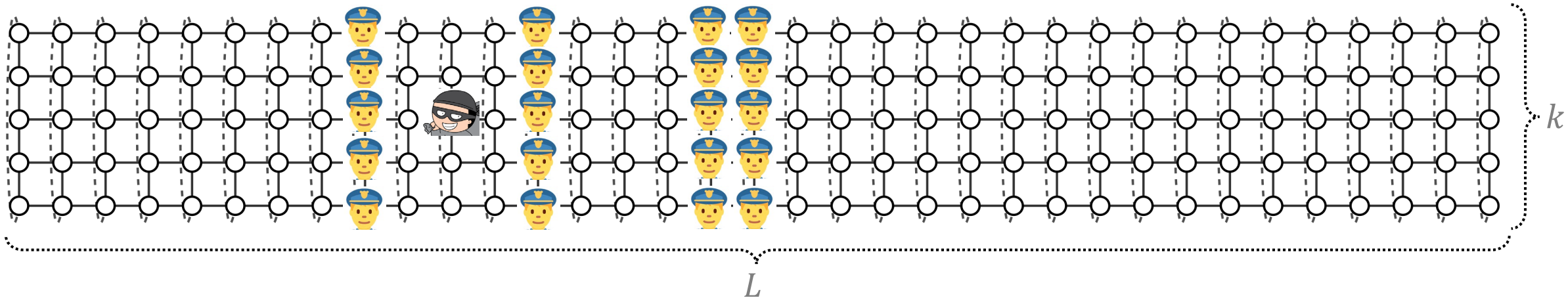


- Start: K cops, one robber at v_0
- In each round:
 - One cop enters helicopter and signal a vertex v
 - Robber moves
 - Cop lands at v
- Ends when robber is caught (by cop at same vertex)

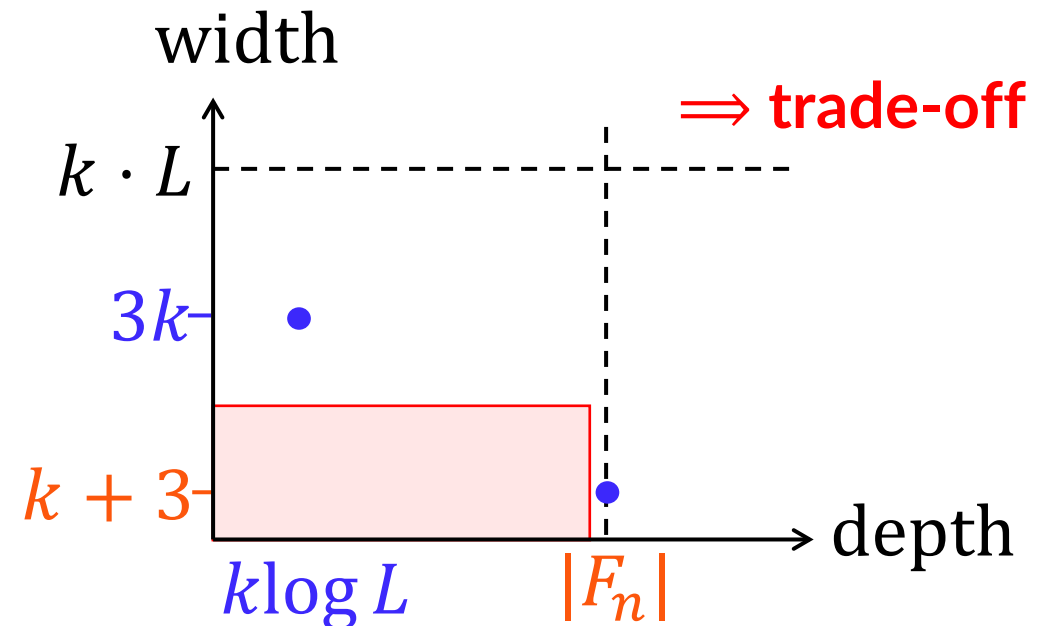
width $\approx$ # cops
depth $\approx$ # rounds

[GTT '18]

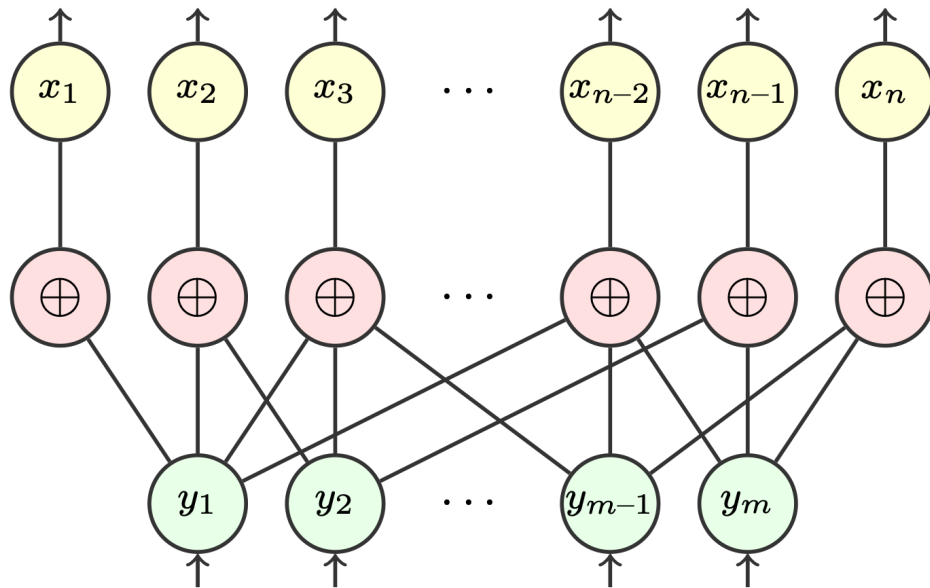
Upper Bounds: Simple Cop Strategies



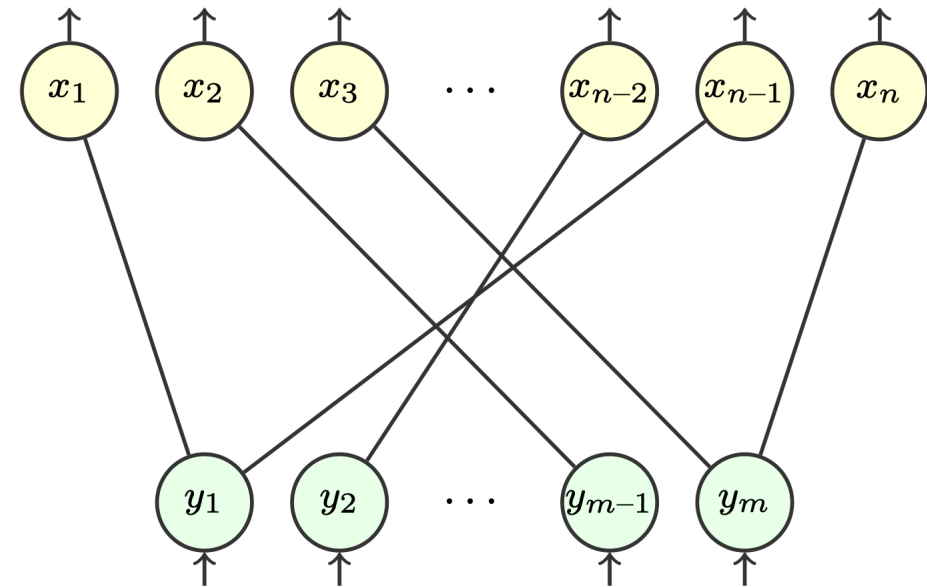
- $k + 1$ cops:
 - Place cops on middle column
 - March towards robber in $k \cdot L$ rounds
 - $\Rightarrow$ translates to resolution proof of width $k + 3$, but depth $k \cdot L$
- $3k$ cops:
 - Binary search
 - $\Rightarrow$ translates to resolution proof of width $3k$ and depth $k \cdot \log L$



Key Idea: Variable Compression [Grohe-Lichter-Neuen-Schweitzer 2023]

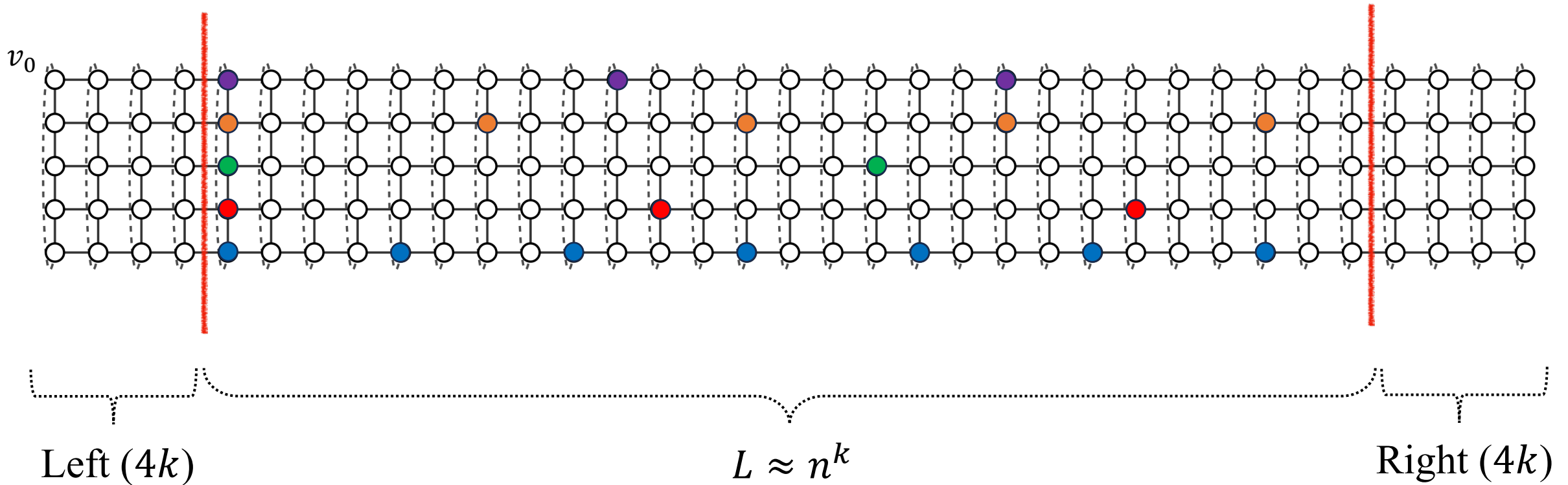


Substitution with XOR gadgets



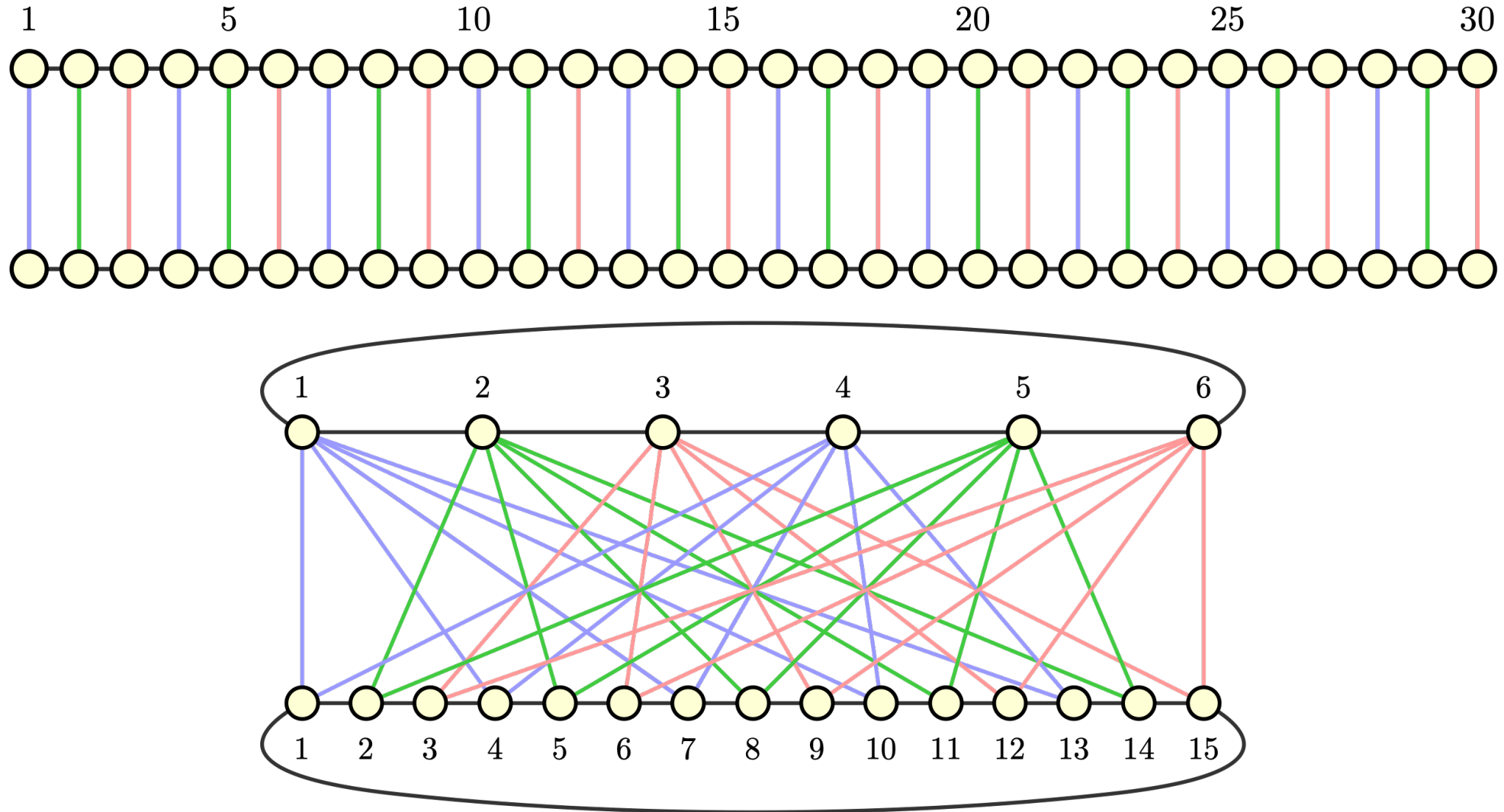
Variable substitution
(with lots of collisions)

Graph and Tseitin Formula after Compression

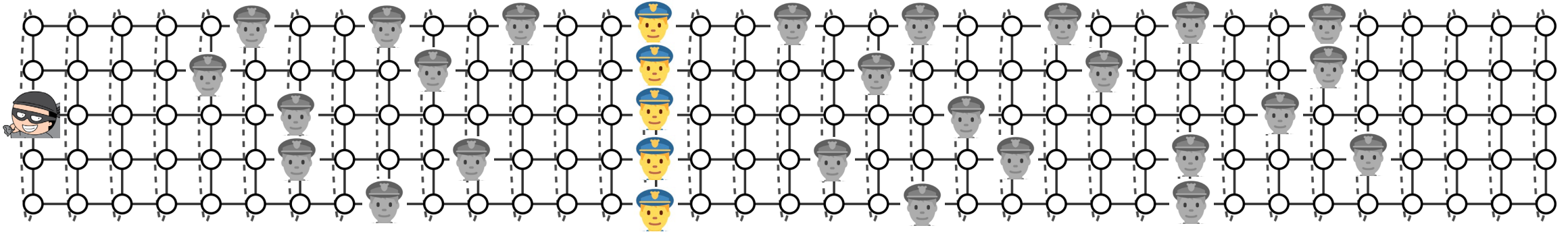


- Vertex equivalence classes $[v_\ell] = \{v_j \mid j \equiv \ell \bmod m_i\}$ for modulus m_i chosen for row i (except ends)
- Induces edge equivalence classes $[e]$
- Compressed formula: $\sum_{[e] \ni [v]} y_{[e]} = 1 \bmod 2$ iff $[v] = [v_0]$

Edge Equivalence Between Two Rows ($m_1 = 6, m_2 = 15$)

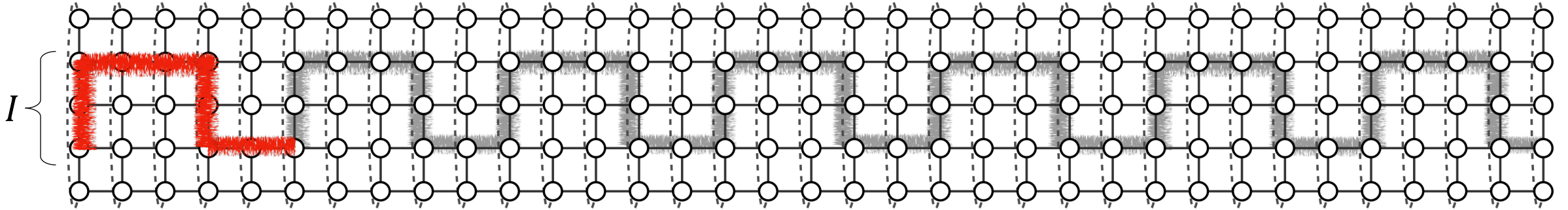


Compressed Game [GLNS '23]



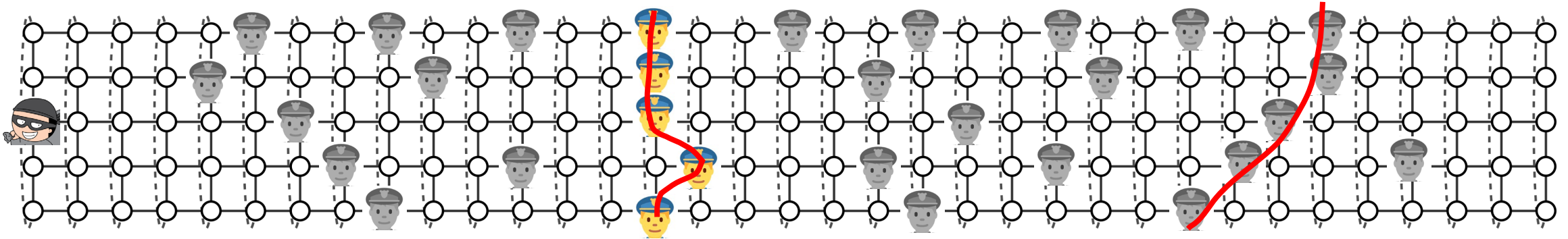
- cop at $v \Rightarrow$ cop at all vertices $u \in [v]$
- $K = k + c$ cops, one robber at v_0
 - Lift a cop and signal a vertex v
 - Robber does a **$\equiv$ -compressible move**
 - Cop lands at $[v]$
- Cop strategies for uncompressed game still work $\Rightarrow$ same upper bounds
- But robber has to avoid cop clones $\Rightarrow$ harder to get lower bounds
- Consider only special type of robber moves

Moves Translatable to Compressed Setting



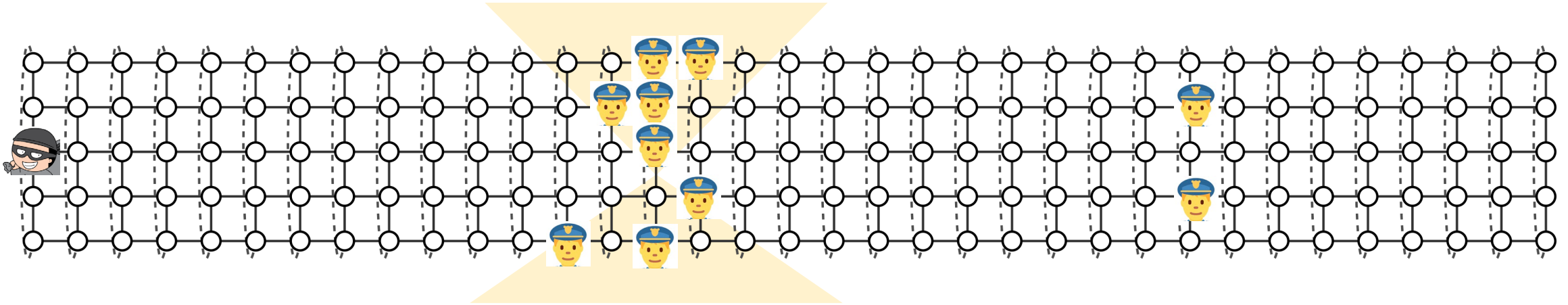
- Robber moves only on $|I| \leq c + 1$ contiguous rows
- Horizontal moves periodic mod $\gcd(m_i: i \in I)$

Idea for Robber Strategy



- Robber stays on cop-free column in left or right end
- Slides between left and right end using compressible moves
- **Problem:** (virtual) cops can form a (virtual) **police cordon**

Idea for Robber Strategy (cont.)

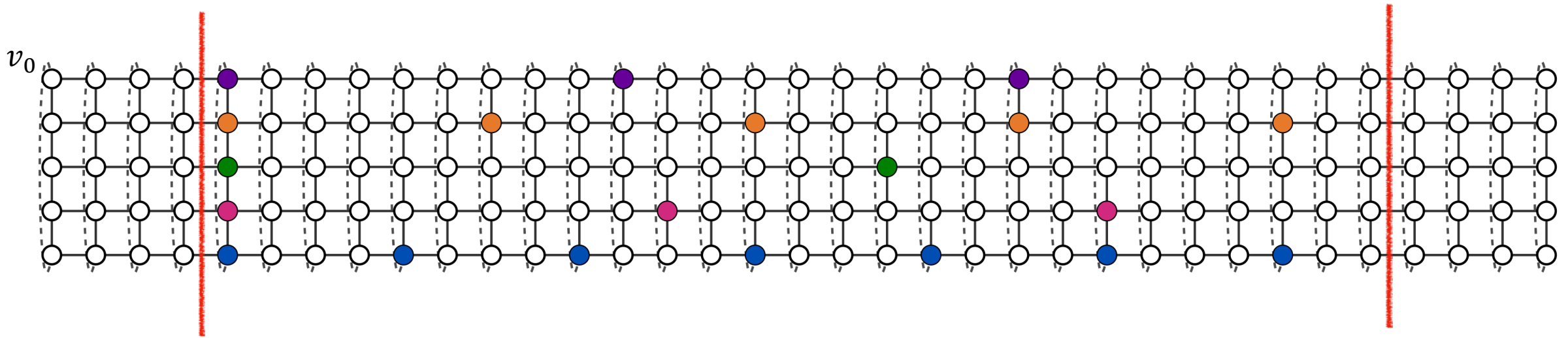


- Plan: Keep away from (set of) **virtual cordons**
- Allows robber to escape in time
- Prove that cordons can only move slowly

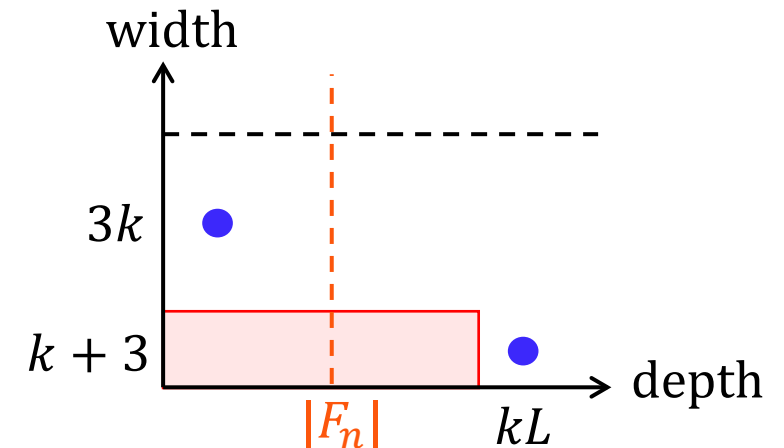


Survives n^k rounds against $k + c$ cops

Choice of Row Moduli



- Fix $1 \leq c \leq k - 2$
 Pick coprime numbers $P_1, \dots, P_k$ where $|P_i| \approx n$
 —*i*th modulus $4k \cdot P_i \cdots P_{i+c}$
 —cylinder length $L \approx 4k \cdot P_1 \cdots P_k$
- Compressed formula size $n^k \rightarrow n^{c+1}$



Motivation for Grohe et al: Weisfeiler-Leman (WL) algorithm

Theorem [Grohe-Lichter-Neuen-Schweitzer '23]

$\exists$ graph pairs k -dimensional WL can distinguish, but only after $n^{k/2}$ iterations.

- Dimension $\approx$ Width
- Iterations $\approx$ Depth
- Graph pair $\approx$ Tseitin formula
- [GLNS '23] not robust enough to yield proof complexity results

Our Result for Weisfeiler-Leman (WL)

Theorem

For any $1 \leq c \leq k - 1$, $\exists$ graph pairs of size n

- dimension- k WL can distinguish
- dimension- $(k + c - 1)$ WL requires $n^{\frac{k}{c+1}}$ iterations

Proof Structure

Proof in two steps:

1. Establish width vs depth trade-off results for resolution
2. Derive proof & circuit complexity trade-off results with lifting

What is a Lifting Theorem?

- Use composition to relate (different) models of computation:
Complexity of f in (weak) model A corresponds to
 $\Rightarrow$ Complexity of composed problem $f \circ g$ in (strong) model B
- Example
 F requires large resolution width
 $\Rightarrow F \circ g$ requires large resolution/cutting planes/monotone circuit size

Lifting with Indexing Gadget (for Functions)

- $f: \{0,1\}^n \rightarrow \{0,1\}$
- Compose with IND: $[m] \times \{0,1\}^m \rightarrow \{0,1\}$, $\text{IND}(x, y) = y_x$

Alice: $x \in [m]^n$

1	3	1	2	2	1	3	2
---	---	---	---	---	---	---	---

Bob: $y \in \{0,1\}^{mn}$

0	0	1	1	0	1	0	1
1	0	0	1	1	0	0	0
0	1	1	1	0	1	0	0

$$f \circ \text{IND}(x, y) = f \left(\begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \hline \end{array} \right)$$

Lifting Resolution to Monotone Circuits [Garg-Göös-Kamath-Sokolov '18]

- Resolution width-depth trade-off $\Rightarrow$ monotone circuit size-depth trade-off

Theorem [GGKS '18]

For CNF formula F and large enough m , can construct function $f_{F,m}$ s.t.
size- m^w , depth- d monotone circuit for $f_{F,m}$
 $\Rightarrow$ width- $O(w)$, depth- $O(dw)$ resolution proof for F

- Crucial issue: #variables of function $\approx$ #clauses in formula
 - Previous supercritical trade-offs do not work
 - Crucial that depth-width trade-off is *truly* supercritical
- Need tighter lifting theorem than [GGKS '18] with better constants

Tight Lifting for Resolution

Theorem

size- $(m/2)^w$, depth- d resolution refutation for $F \circ \text{IND}_m$
 $\Rightarrow$ width- w , depth- d resolution refutation of F

- Essentially optimal
- Simple proof via random restriction

Corollary (size-depth trade-off for resolution)

$\exists$ a CNF formula F on ℓ clauses s.t.

- $\exists$ resolution refutation of F in size s
- But size $\leq \ell s \Rightarrow$ superpolynomial (in ℓ) depth

Tight Lifting for Monotone Circuits

Theorem

size- $m^{(1-\epsilon)w}$, depth- d monotone circuit for $f_{F,m}$
 $\Rightarrow$ width- w , depth- wd resolution refutation of F

- Also almost optimal
- Applies also to cutting planes (and monotone real circuits)

Corollary (size-depth trade-off for monotone circuits)

$\exists$ function f on n variables s.t.

- $\exists$ a monotone circuit for f of size s
- $\text{size} \leq ns \Rightarrow$ superpolynomial depth

Lifting for Tree-like Resolution

Theorem

size- s , width- w tree-like resolution refutation for $F \circ \text{XOR}_{m+1}$
 $\Rightarrow$ depth- $(\log s)$, width- (w/m) resolution refutation of F

- Note that width is also reduced

Corollary (size-width trade-off for tree-like resolution)

$\exists$ a CNF formula F on n variables s.t.

- $\exists$ a tree-like resolution refutation of width $w = \text{poly}(\log(n))$
- But width $\leq w + \sqrt{w} \Rightarrow \exp(\text{superpoly}(|F|))$ size

Concurrent Work [Göös-Maystre-Risse-Sokolov '24]

- Independently obtained supercritical size-depth trade-offs
- Natural formulas formalizing interesting combinatorial principle
- More robust trade-offs
- Can use existing lifting theorems as black box

Our work

- Different parameter regime, with trade-offs kicking in earlier
- Robust supercritical trade-offs also for Weisfeiler–Leman
- Formula compression technique in [GLNS '23] worth investigating further
- Tighter lifting theorems also of independent interest

Some Open Problems

- Further applications of compression in proof complexity
 - Can pebbling formulas be compressed?
 - Can we find other graph compressions?
- Prove more supercritical trade-offs
 - Size vs space or similar
 - Unified parameter range with [\[GMRS '24\]](#)
- Complexity of Tseitin formulas for cutting planes
 - Are there supercritical trade-offs?

Conclusion

- We give *truly* supercritical (in terms of size) trade-offs for
 - proof complexity (resolution and cutting planes)
 - monotone circuits
 - Weisfeiler–Leman algorithm
- Proof in two steps:
 - Base trade-off width vs depth for **compressed** Tseitin formulas
 - Lifting theorems with **improved parameters** yield other results

Future **Supercritical** Research Directions

Thanks for your attention!

- Via **compression** of other formulas or graphs?
- Trade-offs for **other measures** (e.g. size vs space)?
- Size-depth trade-offs for **(uncompressed) Tseitin formulas**?
- Monotone circuit trade-offs for **perfect matching**?